Practice workbook · Calculate a percentage change or percentage difference

Two values of one quantity, from text, a table or a graph, compared as a percentage of a stated starting or reference value.

11 question parts: 1 worked-example retry and 10 further practice parts.
Name: _____ Class: _____ Date: _____

Practise one question type at a time. Attempt the part inside the red boundary; the rest of the question and any figures are kept for context. Write your working in the answer space.

The first question is a retry of the website worked example, without its solution. Check your answers online with 'Check my answer', or ask your teacher for the answer booklet.

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Type ID: HBI-MATH-PERCENTAGE-CHANGE

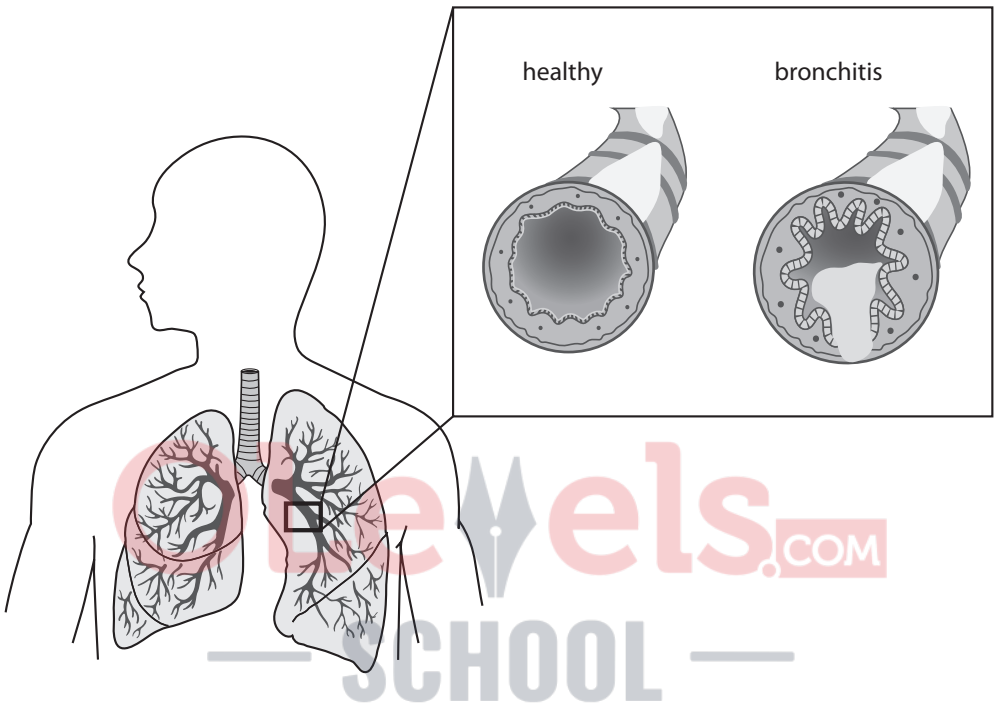
1. Question 6 · part c.ii · 3 marks

January 2020 Paper 1

Worked-example retry · Attempt part c.ii only.

6 Chronic obstructive pulmonary disease (COPD) is a group of lung diseases that includes bronchitis and emphysema.

The diagram shows how bronchitis affects the lungs.



(a) Name the part of the lungs affected by bronchitis.

(1)

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1. Question 6 · part c.ii · 3 marks

January 2020 Paper 1

Worked-example retry · Attempt part c.ii only.

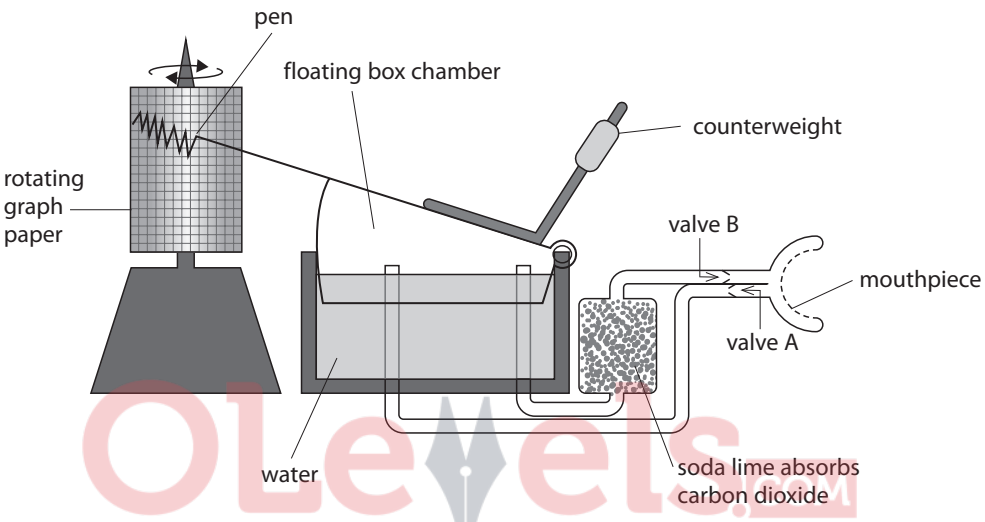
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(b) The diagram shows a spirometer.

A spirometer measures how quickly a person breathes in and out.



(i) Describe how a spirometer is used to obtain a trace of a person's breathing pattern.

(4)

1. Question 6 · part c.ii · 3 marks

January 2020 Paper 1

Worked-example retry · Attempt part c.ii only.

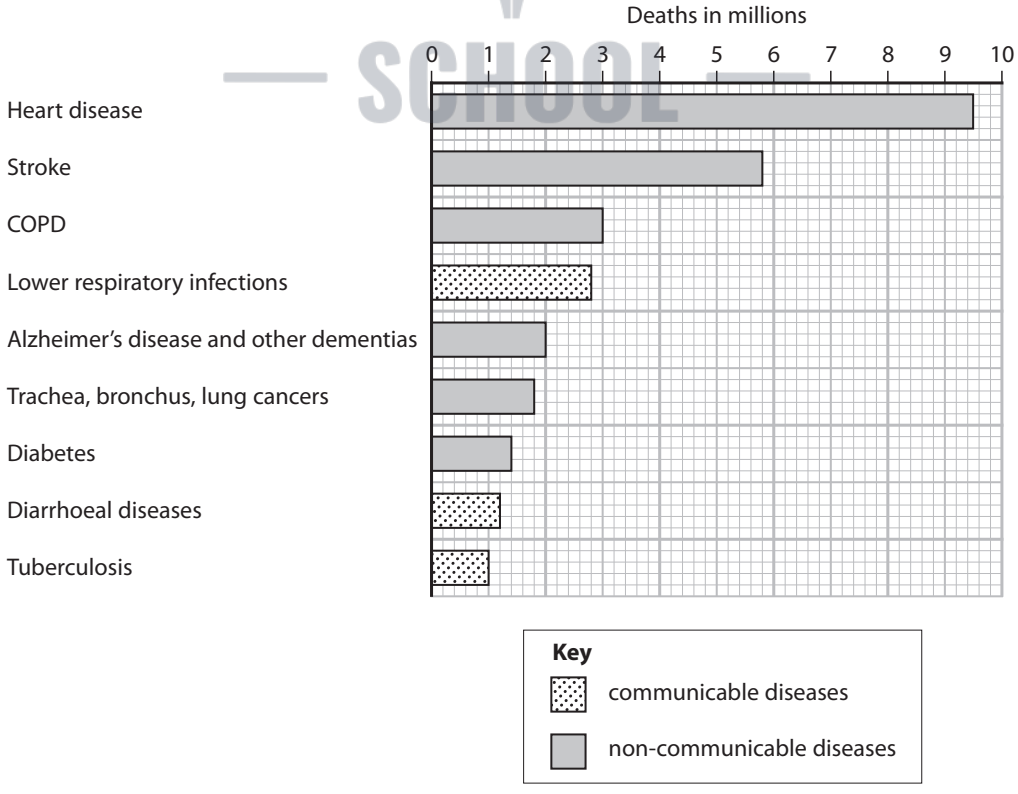
(ii) State one way to reduce risk to health when using a spirometer.

(1)

(iii) Explain why a person with bronchitis needs to take a longer breath to obtain a normal tidal volume.

(2)

(c) The bar chart shows the number of deaths in millions from different health conditions worldwide in 2016.



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1. Question 6 · part c.ii · 3 marks

January 2020 Paper 1

Worked-example retry · Attempt part c.ii only.

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(i) More deaths are caused by non-communicable diseases than by communicable diseases.

Suggest what is meant by the term **non-communicable disease**.

(1)

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(ii) In 2000, the number of deaths from heart disease was 7.0 million.

Calculate the percentage change in the number of deaths from heart disease from 2000 to 2016.

(3)

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percentage change = ...%

(iii) Suggest a reason for the change in the number of deaths from heart disease from 2000 to 2016.

(1)

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(Total for Question 6 = 13 marks)

1. Question 6 · part c.ii · 3 marks · Working and answer

Working and answer

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2. Question 5 · part c.i · 2 marks

June 2019 Paper 2R

Further practice · Attempt part c.i only.

5 Substances move in and out of cells by diffusion, osmosis and active transport.

(a) Describe the process of active transport.

(2)

(b) Water moves in and out of potato cells by osmosis.

Describe a method to determine the concentration of the contents of potato cells.

A large potato is available and also sucrose solutions of various concentrations.

(5)

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2. Question 5 · part c.i · 2 marks

June 2019 Paper 2R

Further practice · Attempt part c.i only.

(c) The table shows results obtained from an investigation to determine the concentration of the contents of potato cells.

	Concentration of sucrose solution in mol per dm³				
	0.2	**0.4**	**0.6**	**0.8**	**1.0**
Initial mass in g	8.4	7.9	8.7	8.3	8.4
Mass after soaking in g	8.7	8.0	8.2	7.5	7.3
Change in mass in g	+ 0.3	+ 0.1	– 0.5	– 0.8	– 1.1
Percentage change in mass	+ 3.6		– 5.8	– 9.6	– 13.1

(i) Calculate the percentage change in mass for 0.4 mol per dm³.

(2)

.....%

(ii) Explain why it is necessary to calculate the percentage change in mass rather than using the change in mass in g.

(2)

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2. Question 5 · part c.i · 2 marks · Working and answer

Working and answer

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3. Question 9 · part a.i · 3 marks

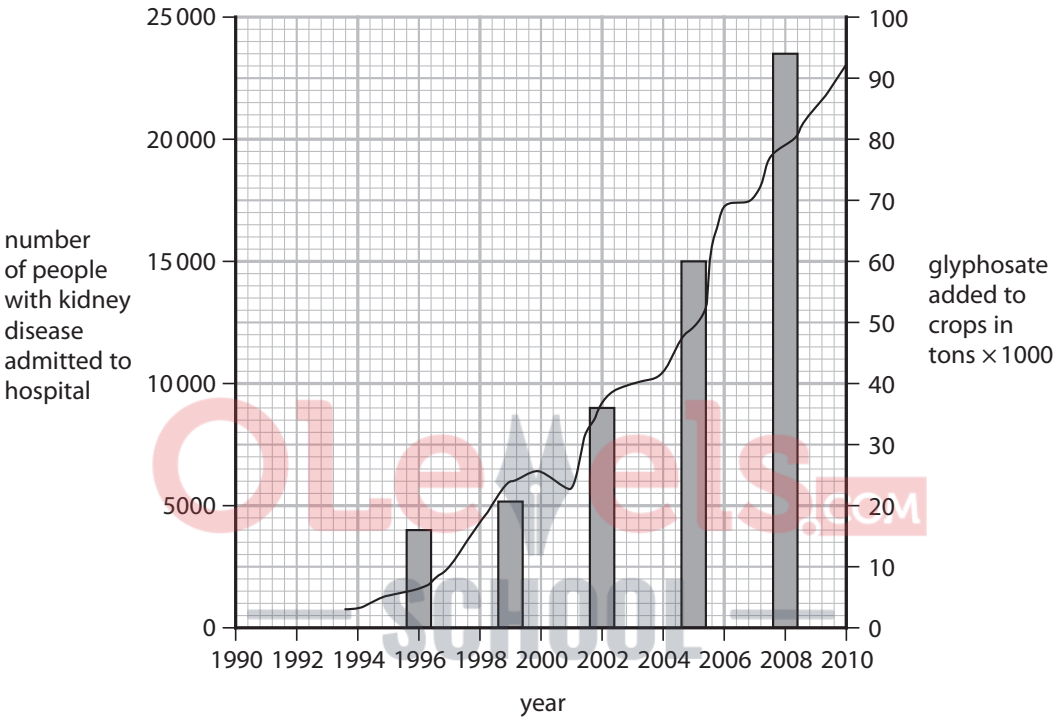
November 2020 Paper 1R

Further practice · Attempt part a.i only.

9 (a) Glyphosate is a chemical found in some weed killers.

The use of this chemical is linked to several health conditions including kidney disease.

The graph shows the number of people with kidney disease admitted to hospital and the amount of glyphosate added to crops.



(i) Calculate the percentage increase in the number of people admitted to hospital with kidney disease in 1996 compared with 2008.

(3)

percentage increase = ... %

20

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3. Question 9 · part a.i · 3 marks

November 2020 Paper 1R

Further practice · Attempt part a.i only.

(ii) Give a reason why the graph does not confirm that glyphosate is a cause of kidney disease.

(1)

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(b) Draw a diagram to show the structure of a kidney nephron.

Add these labels to the diagram.

- glomerulus
- proximal convoluted tubule
- Bowman's capsule

(4)

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3. Question 9 · part a.i · 3 marks · Working and answer

Working and answer

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4. Question 5 · part b.i · 2 marks

November 2020 Paper 2R

Further practice · Attempt part b.i only.

5 A scientist investigates how the drug, atropine, affects the heart rate in humans.

Atropine paralyses the vagus nerve which carries impulses from the brain to the heart.

The heart rates of ten people are measured at intervals of two minutes for eight minutes, after which, atropine is injected.

The scientist continues to measure the heart rates.

The scientist's results are shown in the table.

Time in minutes	Heart rate in beats per minute
0	90
2	81
4	78
6	80
8	82
9	100
10	120
11	148
12	160
13	138
15	128
17	120
19	116
21	109
23	100

atropine injected →

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4. Question 5 · part b.i · 2 marks

November 2020 Paper 2R

Further practice · Attempt part b.i only.

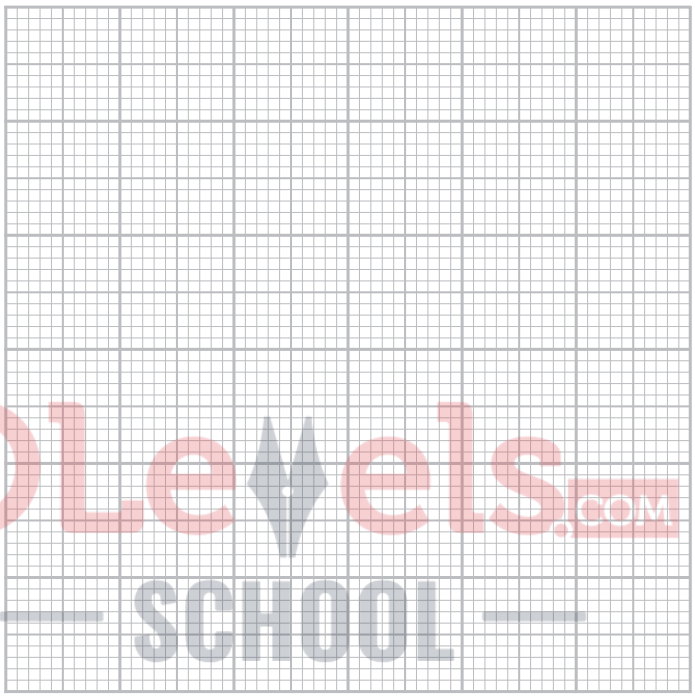
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(a) (i) Plot the scientist's results on the grid. Draw a curve of best fit.

(6)



(ii) Explain why ten people are injected with atropine rather than just one person.

(2)

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(b) (i) Calculate the percentage change in the heart rate at 12 minutes compared to the heart rate at the start of the investigation.

(2)

percentage change = ...%

4. Question 5 · part b.i · 2 marks

November 2020 Paper 2R

Further practice · Attempt part b.i only.

(ii) Explain why this percentage change is not achieved as soon as atropine is injected.

(2)

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(c) Suggest the function of the vagus nerve.

(2)

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(Total for Question 5 = 14 marks)

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Working and answer

5. Question 6 · part b.ii · 3 marks

November 2020 Paper 2

Further practice · Attempt part b.ii only.

6 Read the passage below. Use the information in the passage and your own knowledge to answer the questions that follow.

The effects of alcohol

Humans have been drinking alcohol for thousands of years. Drinking too much alcohol over many years can cause diabetes, malnutrition and diseases of the central nervous system and the liver. One immediate side effect of drinking too much alcohol is increased urination.

Beer is about 95% water and only 5% alcohol. The liver converts that 5% of alcohol into the same volume of water and some carbon dioxide. So if a person drinks 200 cm^3 of beer (one unit), the end result is 200 cm^3 of water. However, this person will not just urinate 200 cm^3 of urine, in fact the person urinates a total of 320 cm^3 of urine. So in general each unit of alcohol makes the person urinate an extra 120 cm^3 of urine more than the normal urine output.

A person with a mass of 60 kg produces 60 cm^3 of urine an hour and a person with a mass of 80 kg produces 80 cm^3 an hour and so on. Therefore if a 60 kg person drinks 200 cm^3 of beer in an hour they will produce 60 cm^3 of urine plus 320 cm^3, making a total of 380 cm^3.

Alcohol interferes with the mechanism that regulates water levels in our body because alcohol affects a small gland called the pituitary gland. One hormone released by the pituitary gland is anti-diuretic hormone (ADH). Alcohol reduces how much ADH is released by the pituitary. Each unit of alcohol that a person drinks forces the kidneys to produce an extra 120 cm^3 of urine more than the normal output.

Logic suggests that drinking lots of water would replace the 120 cm^3 of urine produced, but it is not that simple. The body only retains half the water that a person drinks. The other half leaves the body in urine, so the person will become dehydrated even if they drink lots of water.

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5. Question 6 · part b.ii · 3 marks

November 2020 Paper 2

Further practice · Attempt part b.ii only.

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(a) (i) Where in the body is the pituitary gland (line 16)?

(1)

- ☒ **A** brain
- ☒ **B** kidney
- ☒ **C** liver
- ☒ **D** pancreas

(ii) Which system includes the pituitary gland (line 16)?

(1)

- ☒ **A** digestive
- ☒ **B** endocrine
- ☒ **C** nervous
- ☒ **D** skeletal

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(b) A person weighs 75 kg and drinks 400 cm^3 of beer in an hour (lines 11–14).

(i) Calculate the volume of urine produced by the person in this hour.

(3)

volume of urine = ... cm^3

(ii) Calculate the percentage increase in the volume of urine produced by this person in this hour.

(3)

percentage increase = ...%

5. Question 6 · part b.ii · 3 marks

November 2020 Paper 2

Further practice · Attempt part b.ii only.

(c) Describe the effect of ADH on the kidney (line 17).

(2)

(d) Suggest the effects of dehydration on the blood after increased urination.

(2)

(e) Explain why half of the extra water that the person drinks still leaves the body in the urine (lines 21–24).

(3)

(Total for Question 6 = 15 marks)

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5. Question 6 · part b.ii · 3 marks · Working and answer

Working and answer

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6. Question 4 · part c.i · 3 marks

June 2021 Paper 2

Further practice · Attempt part c.i only.

4 Read the passage below.

Use the information in the passage and your own knowledge to answer the questions that follow.

Obesity and diabetes have been linked to a 20% increase in the number of leg and foot amputations.

Data from Diabetes UK shows that there were 26 378 lower limb amputations related to diabetes from the period 2014 to 2017. This is an increase from the period 2010 to 2013 when there were 22 092 amputations related to diabetes.

Minor lower limb amputations are below the ankle and major lower limb amputations are below the knee. Minor lower limb amputations increased by 26.5% and major lower limb amputations increased by 4.0%.

In 2019 in the United Kingdom 3.8 million people had been diagnosed with diabetes and 90% of these people had Type 2 diabetes.

(a) (i) Describe the causes of obesity.

(3)

(ii) Give two health risks, other than diabetes, associated with obesity.

(2)

1 .....

2 .....

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6. Question 4 · part c.i · 3 marks

June 2021 Paper 2

Further practice · Attempt part c.i only.

(b) Diabetes is caused as a result of insulin not being secreted or not working correctly.

(i) Which organ secretes insulin?

(1)

- ☒ **A** kidney
- ☒ **B** liver
- ☒ **C** pancreas
- ☒ **D** thyroid

(ii) Insulin stimulates the conversion of glucose into glycogen.

In which organ does this occur?

(1)

- ☒ **A** kidney
- ☒ **B** liver
- ☒ **C** pancreas
- ☒ **D** thyroid

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(c) (i) Calculate the percentage increase in the number of lower limb amputations between the period 2010 to 2013 and the period 2014 to 2017.

(3)

percentage increase = ... %

(ii) Calculate the number of people in 2019 who had been diagnosed with Type 2 diabetes in the United Kingdom.

(1)

number of people = ... million

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6. Question 4 · part c.i · 3 marks · Working and answer

Working and answer

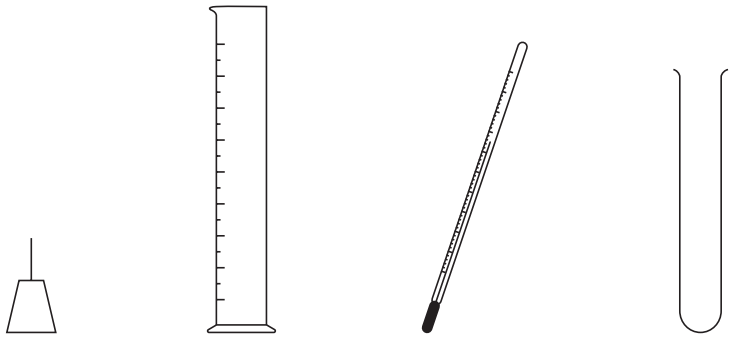
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7. Question 3 · part b.iii · 3 marks

June 2022 Paper 1

Further practice · Attempt part b.iii only.

3 (a) The diagram shows some equipment that can be used to investigate the energy content of different foods.



mounted needle | measuring cylinder | thermometer | test tube

(i) Describe how the equipment can be used to investigate the energy content of different foods.

(5)

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7. Question 3 · part b.iii · 3 marks

June 2022 Paper 1

Further practice · Attempt part b.iii only.

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(ii) Name two variables that need to be controlled in the investigation.

(2)

1 .....

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2 .....

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(iii) Give two safety precautions needed for this investigation.

(2)

1 .....

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2 .....

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7. Question 3 · part b.iii · 3 marks

June 2022 Paper 1

Further practice · Attempt part b.iii only.

(b) Different people have different energy requirements.

The table shows the daily energy requirements in kilocalories (kcal) of males and females of different ages.

Daily Energy Requirement in kcal

Age of adults in years	Males	Females
19–24	2772	2175
25–34	2749	2175
35–44	2629	2103
45–54	2581	2103
55–64	2581	2079
65–74	2342	1912
75 and over	2294	1840

(Source: adapted from https://propranolols.blogspot.com/2020/02/daily-nutritional-requirements-for_9.html)

(i) Describe the patterns in the energy requirements for males and for females.

(2)

(ii) Which of these gives a reason for the difference between the energy required by 25-year-old males and the energy required by 25-year-old females?

(1)

- ☒ **A** females use more muscle mass than males
- ☒ **B** females eat less than males
- ☒ **C** males eat less than females
- ☒ **D** males use more muscle mass than females

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7. Question 3 · part b.iii · 3 marks

June 2022 Paper 1

Further practice · Attempt part b.iii only.

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(iii) Calculate the percentage difference in the energy requirement of males compared with females in the age range of 35–44 years.

(3)

percentage difference = ... %

(Total for Question 3 = 15 marks)

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Working and answer

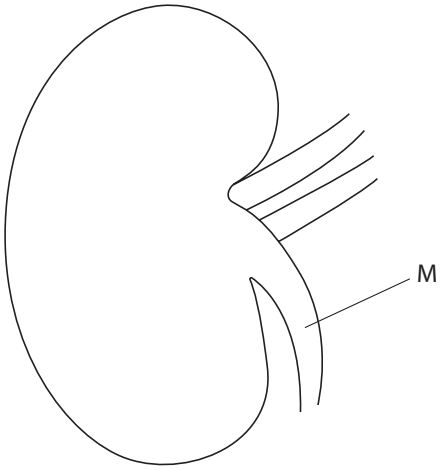
8. Question 7 · part c.i · 3 marks

June 2022 Paper 2R

Further practice · Attempt part c.i only.

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7 The diagram shows a kidney.



(a) (i) Which of these is the name of structure M?

(1)

- ☒ **A** renal artery
- ☒ **B** renal vein
- ☒ **C** ureter
- ☒ **D** urethra

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(ii) Which of these structures is connected to structure M?

(1)

- ☒ **A** bladder
- ☒ **B** prostate gland
- ☒ **C** seminal vesicle
- ☒ **D** urethra

8. Question 7 · part c.i · 3 marks

June 2022 Paper 2R

Further practice · Attempt part c.i only.

(b) State three differences between the blood entering the kidney and the blood leaving the kidney.

(3)

1 .....

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2 .....

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3 .....

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(c) The table compares the nitrogen content of urine during 24 hours for a person on a high protein diet with a person on a low protein diet.

Substance excreted during 24 hours	High protein diet Mass in g	Low protein diet Mass in g
total nitrogen compounds	16.8	3.6
urea	14.7	2.2
uric acid	0.18	0.09
ammonia	0.49	0.04
creatinine	0.58	0.60

(i) Calculate the percentage difference in the mass of nitrogen compounds excreted during 24 hours for the person on a high protein diet compared with the low protein diet.

(3)

percentage difference = ... %

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8. Question 7 · part c.i · 3 marks

June 2022 Paper 2R

Further practice · Attempt part c.i only.

(ii) Which is the main nitrogen containing excretory product produced from the breakdown of protein in the diet?

Use the information in the table to explain your answer.

(2)

(iii) The breakdown of muscle tissue continues at a constant rate and is not affected by diet.

Discuss whether the information in the table supports this statement.

(4)

(Total for Question 7 = 14 marks)

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8. Question 7 · part c.i · 3 marks · Working and answer

Working and answer

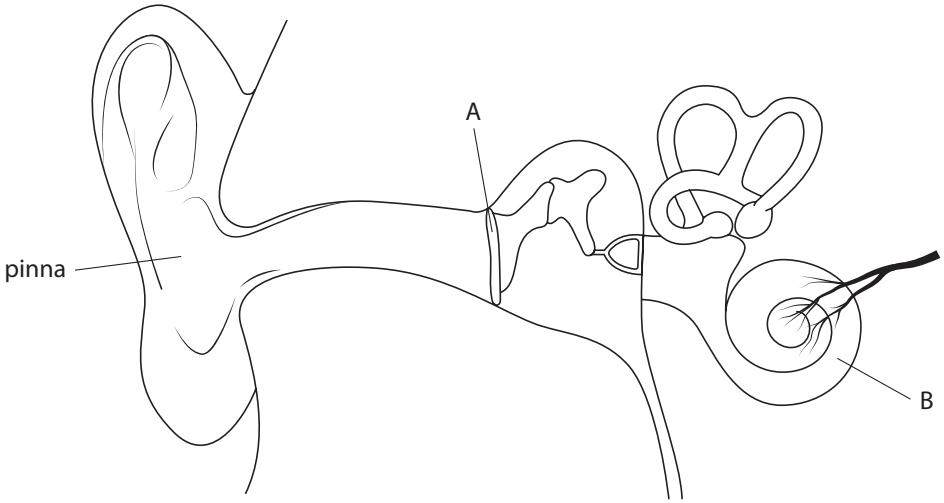
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9. Question 5 · part b.ii · 3 marks

June 2022 Paper 2

Further practice · Attempt part b.ii only.

5 The diagram shows a section through the human ear.



(a) (i) Name parts A and B. **(2)**

A .....

B .....

(ii) Describe the function of the pinna. **(2)**

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9. Question 5 · part b.ii · 3 marks

June 2022 Paper 2

Further practice · Attempt part b.ii only.

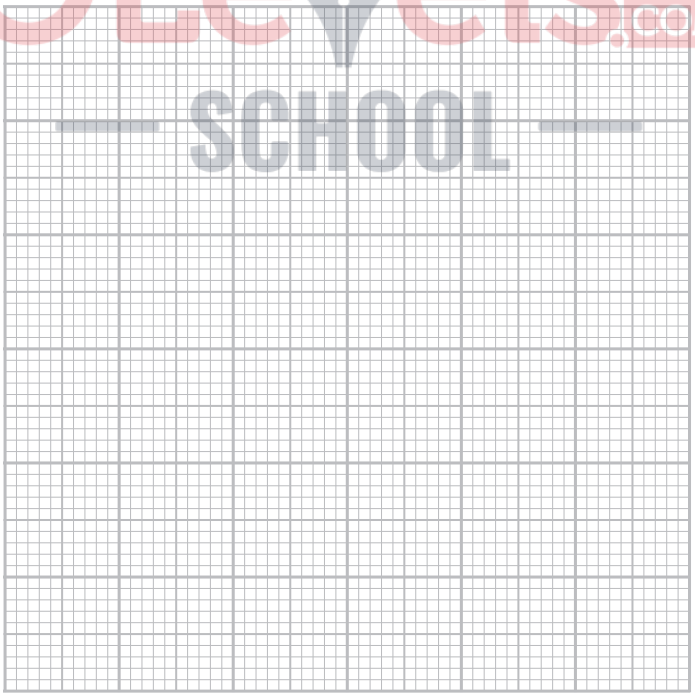
(b) The range of frequencies audible to the human ear can vary with age.

The table shows the maximum frequency detected by people of different ages.

Age in years	Maximum frequency detected in kHz
20	20
30	18
40	16
50	14

(i) Draw a bar chart to show how the maximum frequency detected varies with age.

(4)



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9. Question 5 · part b.ii · 3 marks

June 2022 Paper 2

Further practice · Attempt part b.ii only.

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(ii) Calculate the percentage change in the maximum frequency detected by people aged 20 and people aged 50.

(3)

percentage change = ... %

(c) A student suggests this method that could be used to produce the data shown in the table.

- select a person of each age to be tested
- use a signal generator to produce sounds at different frequencies
- record the maximum frequency detected by each person

(i) Give two reasons why this is unlikely to produce valid results.

(2)

1 .....

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2 .....

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(ii) Describe how the reliability of the results could be improved.

(2)

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(Total for Question 5 = 15 marks)

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9. Question 5 · part b.ii · 3 marks · Working and answer

Working and answer

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10. Question 8 · part b.ii · 2 marks

June 2023 Paper 2R

Further practice · Attempt part b.ii only.

8 (a) (i) Describe the structure of a virus.

(2)

(ii) Viruses attach to cells of various body tissues.

The virus uses the host cell's genetic material to produce new viral proteins.

Describe how the host cell carries out protein synthesis to produce new viral proteins.

(5)

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10. Question 8 · part b.ii · 2 marks

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Further practice · Attempt part b.ii only.

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(b) In 2010, AIDS-related illnesses caused the death of 1.4 million people.

In 2021, AIDS-related illnesses caused the death of 650 000 people.

(i) Name the microorganism that causes AIDS.

(1)

.....

(ii) Calculate the percentage decrease in the number of people dying from AIDS-related illnesses between 2010 and 2020.

(2)

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percentage decrease = ... %

(Total for Question 8 = 10 marks)

Working and answer

11. Question 3 · part a.ii · 3 marks

November 2023 Paper 1

Further practice · Attempt part a.ii only.

3 The table shows the flow of blood to different organs in a person at rest and during exercise.

	Blood flow in cm^3 per minute	
	at rest	during exercise
Heart	250	750
Skeletal muscles	1000	12 500
Skin	400	1900
Gut	1400	600
Brain	750	750
Total flow	5600	17 500

(a) (i) Give the ratio of the blood flow to the heart when at rest and during exercise.

Give your answer in the form 1 : n

(2)

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Ratio = ... : .....

(ii) Calculate the percentage increase in the total flow of blood between rest and exercise.

Give your answer to two significant figures.

(3)

percentage increase = ... %

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11. Question 3 · part a.ii · 3 marks

November 2023 Paper 1

Further practice · Attempt part a.ii only.

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(iii) Explain why the total blood flow at rest is greater than the flow to the heart, skeletal muscle, skin, gut and brain added together.

(3)

(b) (i) Describe the main changes between the blood flow values at rest and during exercise.

(3)

(ii) Explain the change in blood flow rate to the skin when at rest and during exercise.

(4)

(Total for Question 3 = 15 marks)

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11. Question 3 · part a.ii · 3 marks · Working and answer

Working and answer

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